CM3020 Artificial Intelligence

Midterm Part B

Word Count:1927

1. Baseline

1.1 Redesign Fitness Function

After reviewing cw-envt.py and instantiating the arena–mountain scene, attention turned to formulating a fitness function that rewards agents for reaching the peak's coordinate. The peak's coordinate is obtained with getAABB(), which returns an axis-aligned bounding box, extracting peak's z-coordinate. The Euclidean distance between the creature's centre of mass and the peak is calculated at every time-step, and fitness is set to

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where is the arena's longest possible diagonal. Hence, the score rises as the agent approaches the peak, equals when the peak is reached, and falls to if the agent leaves the arena. Distance alone was chosen for its direct, task-specific reflection of climbing success.

1.2 Experiment on Different Settings

With the fitness function established, three evolutionary hyperparameters were probed: population size (10 vs 20), initial genome length (2 vs 3 nodes), and point-mutation rate (0.1 vs 1.0). And for each run, three metrics were logged: mean population fitness, best-of-run fitness, and maximum link count.

Population size was varied to determine whether a larger breeding pool accelerates or stabilizes adaptation. Initial genome length was adjusted to test whether agents that begin with fewer behavioral primitives can still evolve into sufficiently complex morphologies. Finally, the point-mutation rate was altered to measure the impact of random genetic changes on overall performance.

Each run was restricted to 300 generations, a threshold that prevented the "GetBasePositionAndOrientation failed" error occasionally encountered in longer PyBullet sessions, an issue attributed to accumulating computational load. This limit preserved stability while still allowing ample evolutionary progress.

	A	B	C	D	E	F	G	H	I
1	Run ID	Population Size	Initial Genome Size	Point Mutation Rate	Number of Generations	Average Fitness	Best Fitness	Max Links	Note
2	1	10	3	0.1	300	7.881	8.211	124	-
3	2	20	3	0.1	300	7.844	8.906	121	Cheated; flying into the air
4	3	10	2	0.1	300	7.437	8.156	121	-
5	4	20	2	0.1	300	8.042	8.3	43	-
6	5	10	3	1	300	7.379	7.553	67	Cheated; flying into the air
7	6	20	3	1	300	6.901	9.181	122	-
8	7	10	2	1	300	6.908	7.609	121	-
9	8	20	2	1	300	6.908	7.091	76	-

The table summarizes the eight experimental runs and reveals several patterns. First, the difference between populations of 10 and 20 is little, both average- and best-fitness scores are nearly identical, indicating that population pressure in this range neither hastens nor hinders adaptation. Although larger populations seemed to lengthen run-time, execution time was not systematically logged, so this remains a qualitative observation rather than a quantified effect.

Second, agents seeded with two-node genomes rapidly added links and reached fitness levels indistinguishable from those of three-node starters, demonstrating that a minimal initial encoding does not impede the evolution of climbing-capable complexity.

Third, increasing the point-mutation rate from 0.1 to 1.0 sharply lowered fitness and compressed the gap between mean and best scores. At 1.0, continual gene re-writes erase emerging adaptations before selection can reinforce them, whereas a 0.1 rate lets advantageous motifs persist and compound, yielding both higher peak performance and a wider elite–median spread.

population size and initial genome length exert only secondary effects. Excluding the cheating cases in Runs 2 and 5, Runs 1 and 4 produced the highest legitimate fitness. To minimize computation time, the study adopts the Run 1 settings, population = 10, genome length = 2, mutation rate = 0.1, for all subsequent experiments.

1.3 Summary

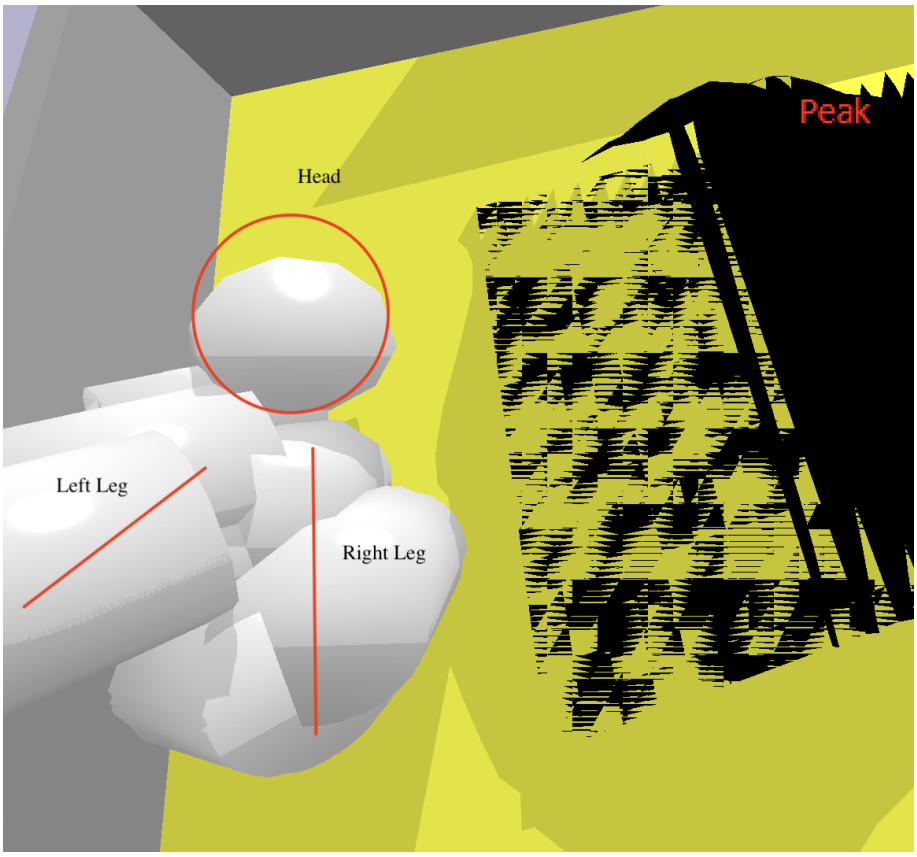
Replay analysis in realtime_from_csv.py showed that several agents gamed the distance-only metric: they enlarged their bodies until a limb lay near the peak, thus posting high fitness without climbing. Some even remained stationary, or drifted away from the mountain, yet still registered competitive scores.

These observations expose a flaw in the distance-only metric: because fitness depends solely on Euclidean gap to the peak, selection rewards any tactic that shrinks this distance, regardless of behavioral plausibility.

To close this loophole, the next section augments the fitness function with a penalty for excessive body volume and a reward for sustained vertical gain. These additions are expected to steer evolution toward genuine climbing behavior while suppressing growth-only exploits.

1.4 Additional Findings

An interesting morphology emerged in Run 7 (elites_299). The final elite individual assumed a configuration reminiscent of a human figure mid-stride, although the body lay prone on the ground. This outcome illustrates one of the compelling qualities of genetic algorithms: the system's inherent stochasticity can yield solutions whose forms and behaviors defy prior expectation, revealing an almost artistic spontaneity within the evolutionary search.



2. Exceptional Section 1 - Improve Fitness Function

2.1 Redesign Fitness Function

The updated fitness metric augments simple distance-to-peak with three elements. A negative weight is applied to the creature's total volume, disqualifying individuals relying on unrestrained growth. Net vertical gain is rewarded by adding the height difference between successive timesteps, steering evolution toward genuine ascent. Milestone bonuses are issued at 75 % and 100 % of mountain height, thresholds reachable only through sustained climbing rather than size inflation, thereby reinforcing legitimate upward progress.

2.2 Experiment on Different Settings

To calibrate the revised fitness function, the study systematically varied the penalties assigned to body size and the rewards granted for upward movement. Size weights of −0.3, −0.5, and −0.8 were paired factorially with upward-movement weights of 0.3, 0.5, and 0.8, producing nine distinct parameter combinations. Each simulation run was assessed using the minimum Euclidean distance the agent achieved relative to the peak, providing a consistent benchmark for comparing the effectiveness of the different weight settings.

	A	B	C	D	E
1	Run ID	Size Weight	Upward Movement Weight	Distance to Peak	Note
2	9	-0.3	0.3	10.57	-
3	10	-0.5	0.3	10.71	-
4	11	-0.8	0.3	10.802	-
5	12	-0.3	0.5	9.592	Kind of cheating with Upward Movement score; Waving hand
6	13	-0.5	0.5	10.986	Kind of cheating with Upward Movement score; Flipping itself like a coin
7	14	-0.8	0.5	10.824	Kind of cheating with Upward Movement score; Waving hand
8	15	-0.3	0.8	9.303	-
9	16	-0.5	0.8	10.571	Kind of cheating with Upward Movement score; Waving hand
10	17	-0.8	0.8	8.771	Shape like a skier; Kind of cheating with Upward Movement score; Moving up/down

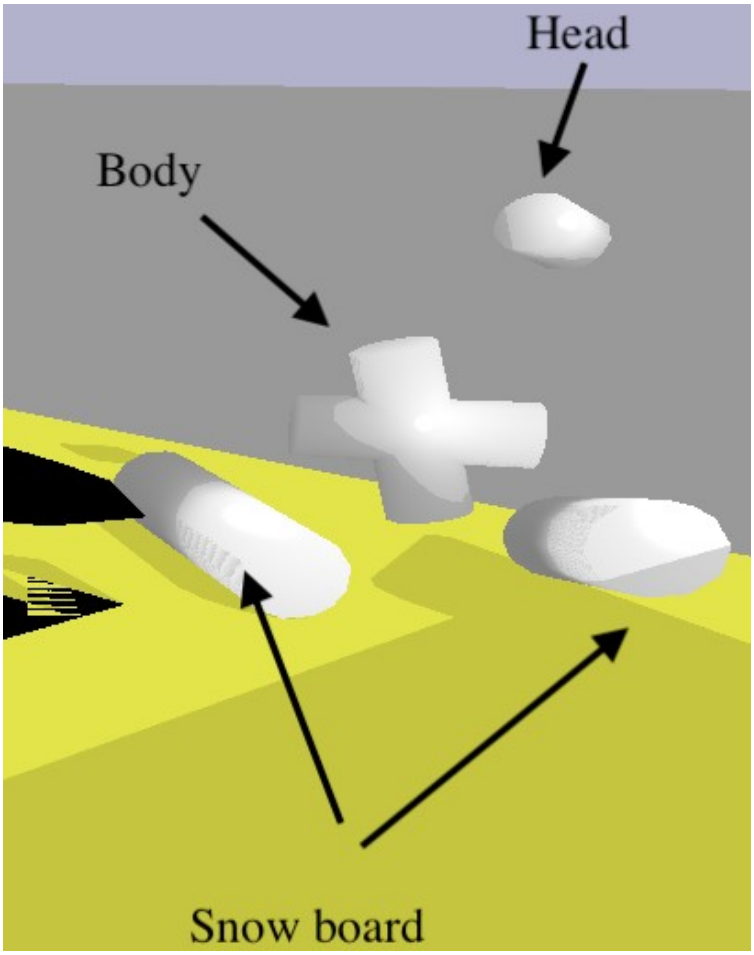
The results reveal that most runs still finished with large peak-to-body gaps. Post-run inspection shows why: in all but Runs 13 and 17 the elite agents evolved pendulum-like limbs that swung in place, producing momentary height spikes that repeatedly triggered the vertical-gain reward and inflated fitness even though the centre of mass made no real progress toward the peak.

2.3 Summary

Agents continued to exploit reward loopholes, and the expanded fitness metric still failed to produce consistent climbing. Runs 13 and 15, for instance, generated individuals that wandered aimlessly, underscoring the weakness of distance-based rewards. The next phase will enclose the spawn zone with containment walls, leaving the mountain as the only exit. By making upward traversal the sole escape route, the revised arena should steer evolution toward authentic ascent and suppress further gaming.

2.4 Additional Findings

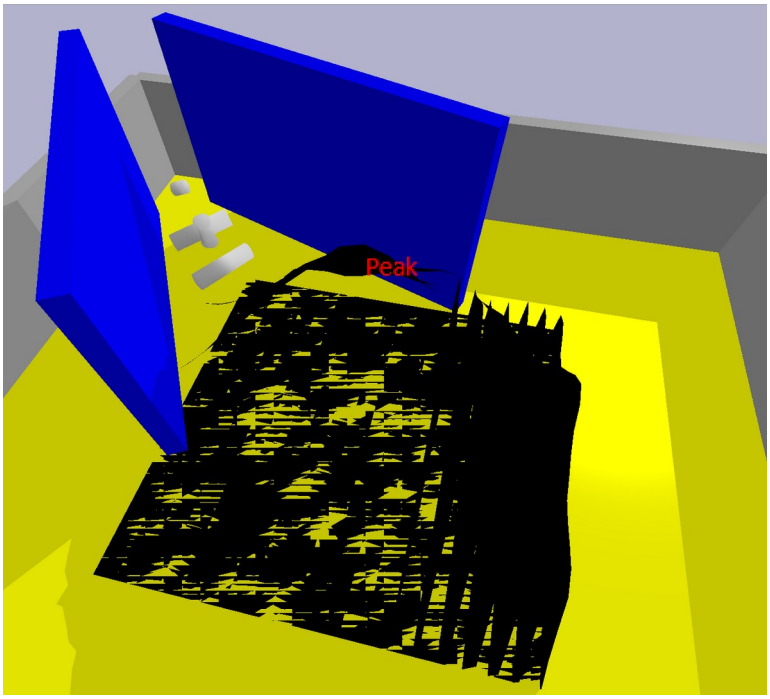
Run 17 (elites_299) yielded a particularly distinctive morphology: the agent evolved a head-and-torso structure flanked by two broad, board-like limbs, evoking the silhouette of a downhill skier. Despite this visually coherent form, the “snowboards” never lifted far enough from the surface to generate the clearance required for upward propulsion, and the creature ultimately failed to initiate a climb.



3. Exceptional Section 2 - Different Landscapes

3.1 Implementation & Testing

The experimental landscape was further modified by erecting two walls that enclosed the respawn zone, thereby creating a more constrained arena. This initial barrier arrangement (figure below) was intended to channel the agents' movement toward the mountain and to observe how the added spatial restriction would influence their climbing strategies.



Early generations often drifted toward the walls, but later cohorts adjusted their paths and moved toward the mountain, indicating that the added constraints were beginning to steer agents along more purposeful trajectories.

Over longer runs, however, the earlier exploitative behaviours resurfaced. Two patterns were especially persistent: some agents ballooned in size to fake proximity, while others launched themselves skyward to score brief distance gains.

To suppress the first tactic, the size-penalty coefficient was raised in stages from −0.3 to −0.8, yet the larger weight failed to reduce the prevalence of "growth-only" solutions. Examination of the reward trace revealed why: even heavily penalised giants continued to collect substantial milestone bonuses, leaving them competitive enough to survive selection. This oversight echoed the weakness noted in Section 2, where the bonus terms had been introduced without considering their vulnerability to exploitation.

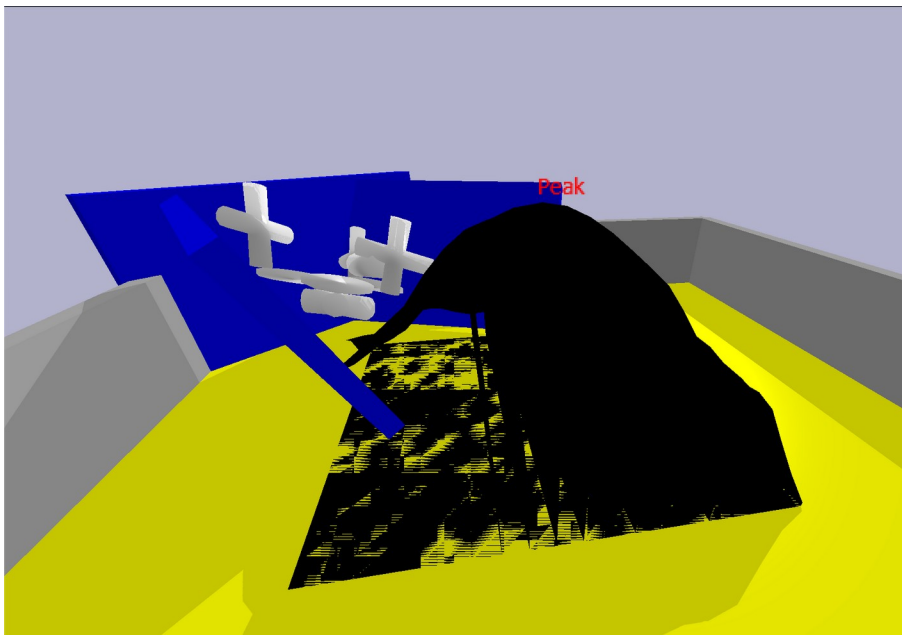
To verify the hypothesis, all bonus components were removed and the simulation was restarted with the original size-weight value of −0.3. Eliminating the extra rewards immediately curtailed the dominance of oversized morphologies, confirming that the bonuses, rather than the penalty magnitude, were the principal enablers of the cheating strategy.

The aerial exploit traced to arena geometry: the narrow walls occasionally redirected horizontal forces upward, turning them into launch ramps. The next iteration widens and tilts these barriers to close this loophole.

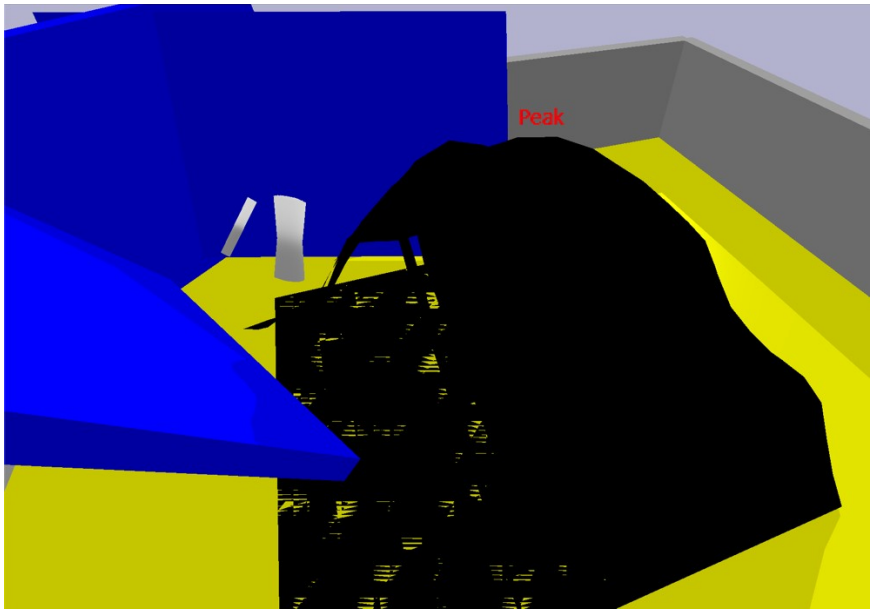
3.2 Implement New Landscape

The barriers were tilted with the baseOrientation parameter in createMultiBody, giving each wall a slight lean rather than a perpendicular stance. This rotation created extra clearance, allowing new agents to settle onto the terrain instead of being flung outward. A third wall was then added behind the spawn zone, leaving the mountain as the only exit and further steering agents toward the slope.

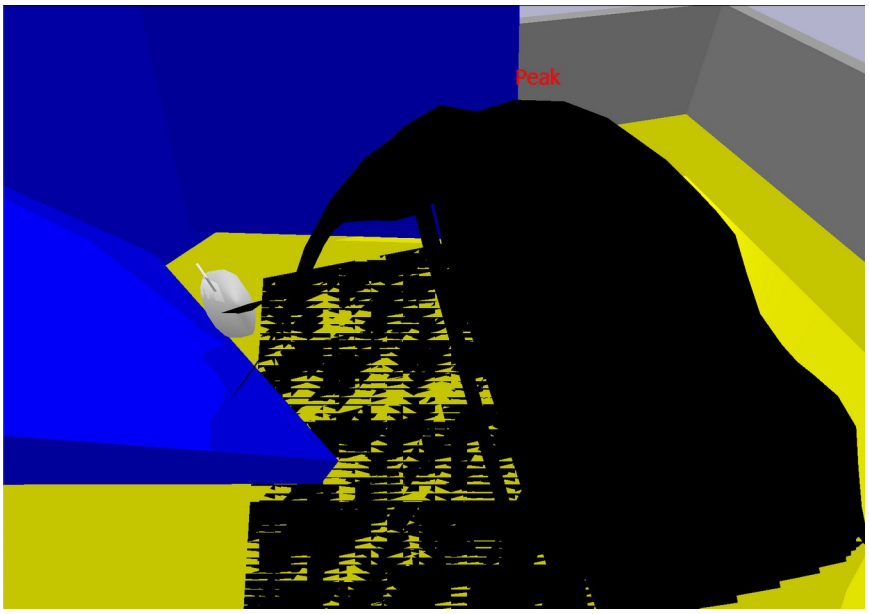
Despite these adjustments, repeated simulation runs produced a familiar pattern: by the final generation, elite individuals tended either to be oversized, still exploiting the distance metric through sheer bulk, or so small that they lacked the leverage to ascend. Representative morphologies from the last generation are presented below to illustrate this bimodal outcome.



2nd Version – Size Too Big



5th Version – Size Too Small



8th Version – Size Too Small

3.3 Summary

Despite extensive trials, no agent ultimately reached the peak; however, the experiments yielded valuable lessons on genetic-algorithm behavior. Across both original and modified arenas, creatures devised inventive, sometimes exploitative, strategies that underscored the tight coupling between reward design, environmental constraints, and emergent behavior. These findings highlight the necessity of carefully balanced fitness signals. Future work will investigate alternative genotype encoding schemes and vary motor-actuation patterns to assess their influence on morphology and climbing performance.

4. Advanced Section – Different Motor Types

4.1 Implementation & Testing

A follow-up experiment tested whether restricting each motor to a single actuation waveform would change performance. Alongside the starter code's pulse and sine patterns, two new waveforms were added: a sawtooth ramp that resets after each linear rise and a noise signal that samples a random value between −1 and 1 every timestep. The simulation was rerun with all four patterns to compare their effects on best and mean fitness.

	A	B	C	D
1	Motor Types	Best Fitness Score (using parameters of fitness function RunID 9)	Mean Fitness Score	Note
2	Pulse	7.185	6.654	Cheat in size. Higher momentum
3	Sine	8.034	7.586	Cheat in size. Smoother and lower momentum
4	Sawtooth	5.267	4.637	Smoother and lower momentum
5	Noise	4.982	3.366	Impossible to predict movement

The persistent bimodal morphologies signal that fitness alone remains an unreliable performance gauge, yet the final populations still clarify how each actuation pattern shapes behavior. The pulse waveform's rapid swings impart high momentum, fostering more vigorous motion. Sine and sawtooth signals, lacking this impulse, produce smoother but slower trajectories. Noise actuation yields erratic dynamics; the wide gap between mean and best fitness indicates that evolution seldom discovers productive uses for this stochastic motor.

4.2 Summary

Sine and sawtooth waveforms generated smoother, more natural motion but lacked the impulse required for consistent ascent; they would suit experiments that prioritize biological realism over raw force. The stochastic noise signal proved unusable, offering no stable gradient and leaving evolution unable to integrate it into productive behavior. Pulse actuation emerged as the most effective choice, its high-momentum swings supplying the thrust needed for uphill propulsion in this task.

5. Summary

This study set out to evolve a virtual creature capable of scaling a PyBullet-generated mountain, iteratively redesigning both the fitness function and the environment. A purely distance-based reward proved vulnerable to two exploits, size inflation and ballistic leaps, prompting the introduction of penalties for body volume, rewards for net vertical gain, and milestone bonuses. Although these additions reduced some pathologies, agents continued to game residual loopholes, demonstrating how tightly evolutionary outcomes depend on reward specification. Confinement walls and later wall tilts channeled movement toward the slope but did not eliminate cheating, ultimately yielding a bimodal population of oversized “balloons” and undersized, ineffective morphologies. Mutation pressure emerged as the principal driver of adaptation, while population size and initial genome length showed minimal impact. Among actuator patterns, pulse motors delivered the highest momentum and thus the best, albeit still inadequate, climbing attempts; sine and sawtooth produced smoother but weaker motion, whereas noise signals remained ineffective. No agent reached the peak, yet the experiments underscored the necessity of carefully balanced fitness signals and thoughtful landscape design. Future work will examine alternate genotype encodings and motor schemes that may foster more robust, cheat-resistant climbing strategies.